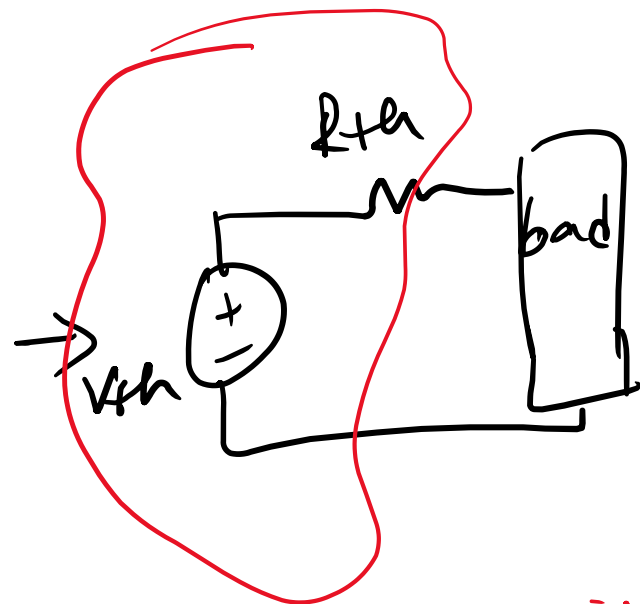
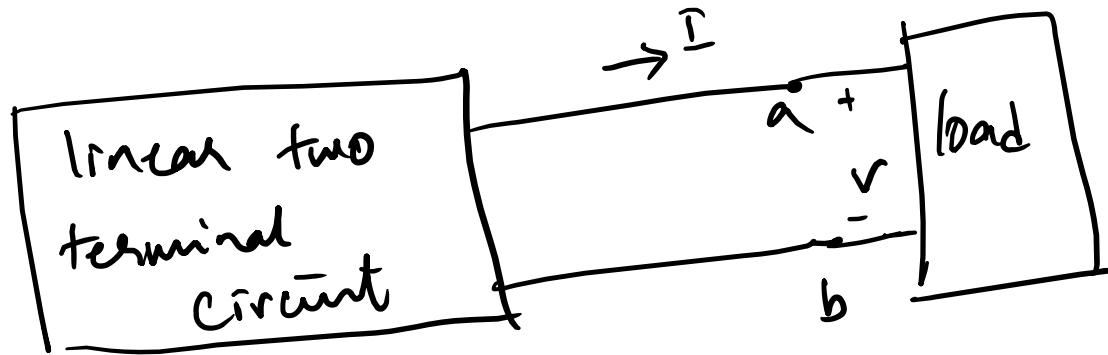


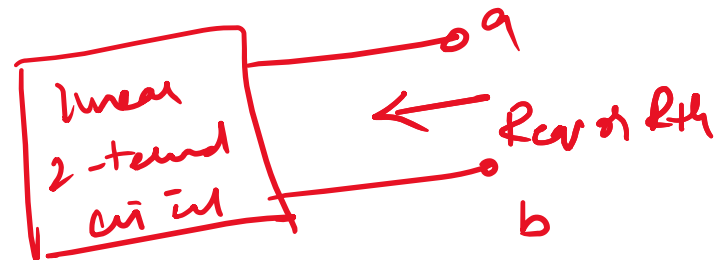
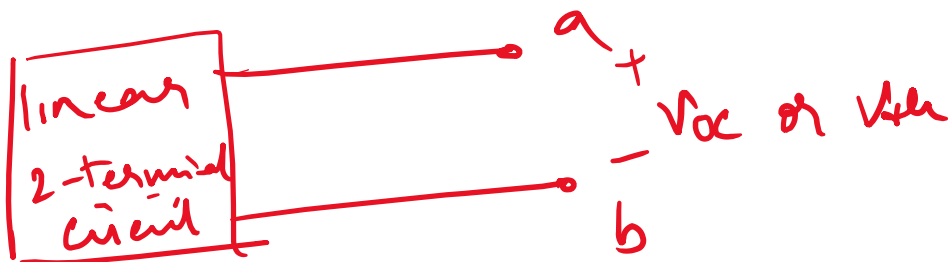
Thevenin's theorem



V_{th} : Thevenin voltage or open circuit voltage at the terminals.

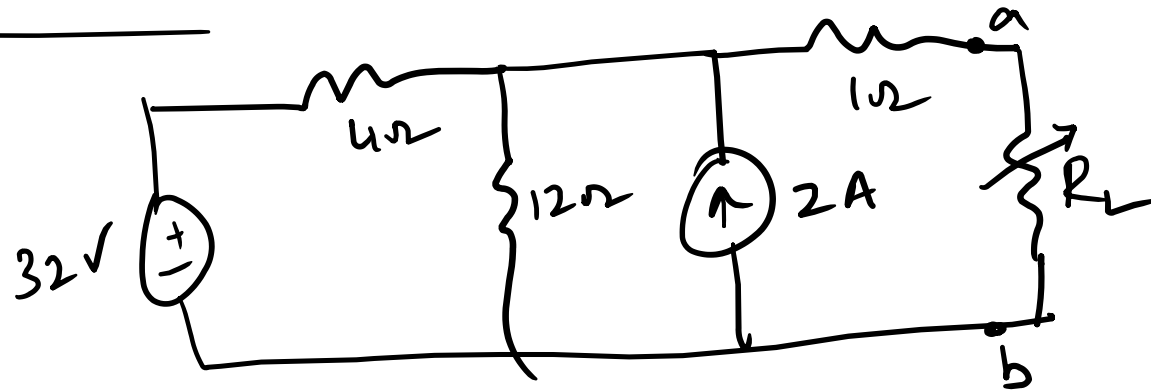
Thevenin equivalent circuit.

R_{th} : Thevenin resistance or input or equivalent resistance at the terminals when the independent sources are turned off.



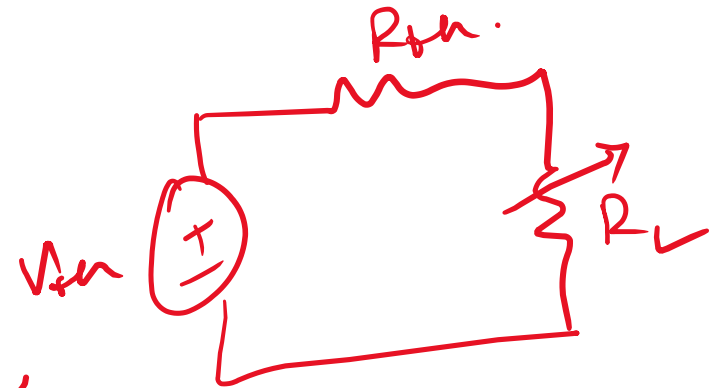
Ind Sources off: voltage source $\rightarrow$ short circuit

Current source $\rightarrow$ open circuit.

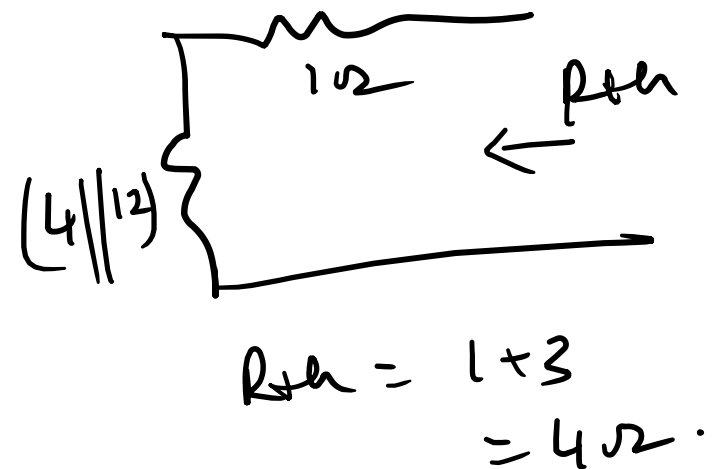
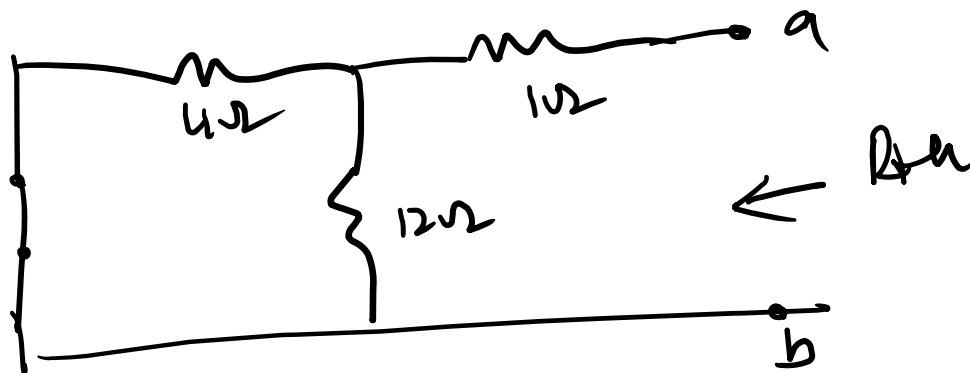


Find the Thevenin equivalent circuit at a-b

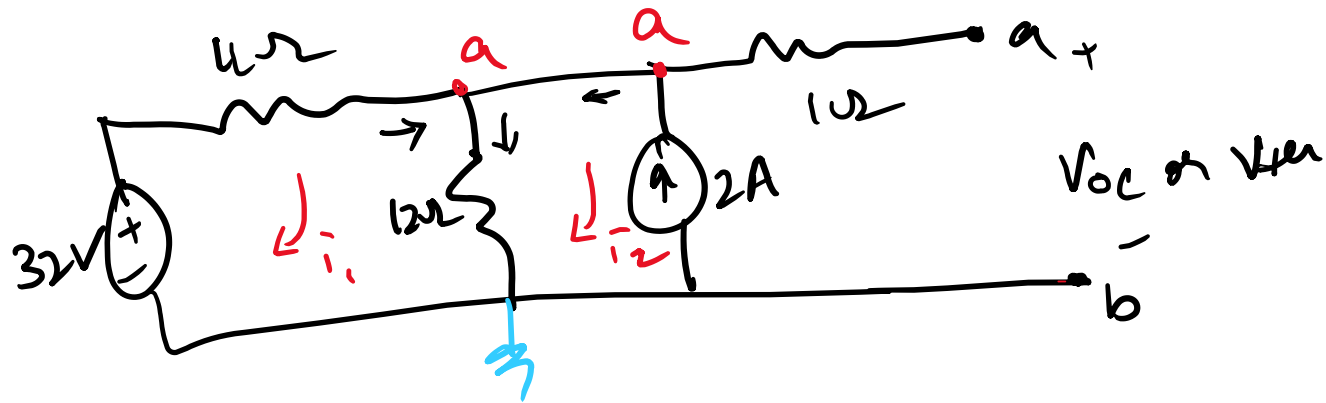
Find V_{th} , R_{th} .



1) Find R_{th} : Turn off all ind sources, remove load, measure R equivalent looking inward from load terminals.



2) V_{th} & V_{oc} : Remove load, find open circuit voltage at load terminals.



$$\hat{i}_2 = -2A$$

$$32 - 4\hat{i}_1 - 12(\hat{i}_1 - \hat{i}_2) = 0$$

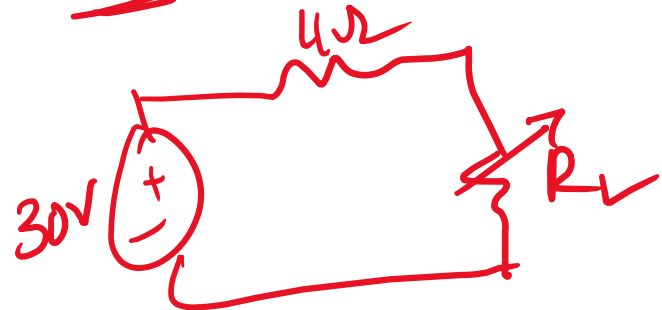
$$\hat{i}_{12\Omega} = \hat{i}_1 - \hat{i}_2$$

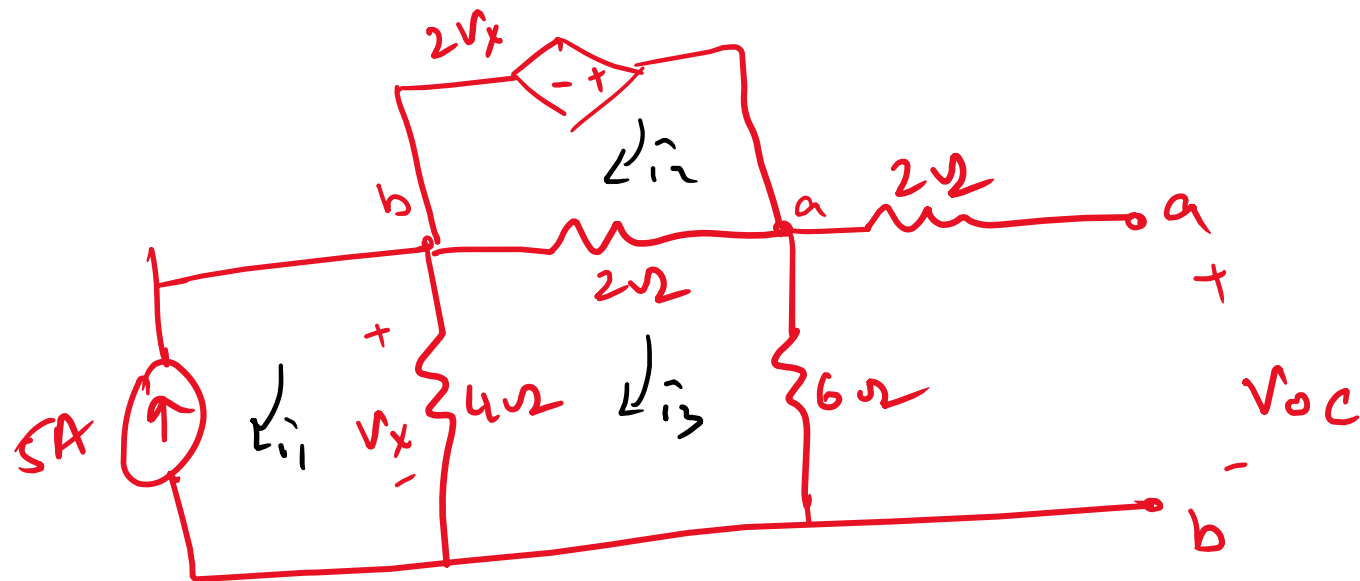
$$\hat{i}_1 = 0.5A$$

$$V_{12} = 12(0.5 - (-2)) = \underline{\underline{30V}}$$

$$\frac{32 - V_a}{4} + 2 = \frac{V_a}{12}$$

$$V_a = \underline{\underline{30V}}$$





Find Thevenin equivalent.

KVL :

$$-2(\hat{i}_2 - \hat{i}_3) + 2v_x = 0$$

$$-4(\hat{i}_3 - \hat{i}_1) - 2(\hat{i}_3 - \hat{i}_2) - 6\hat{i}_3 = 0$$

$$\hat{i}_2 = 10/3$$

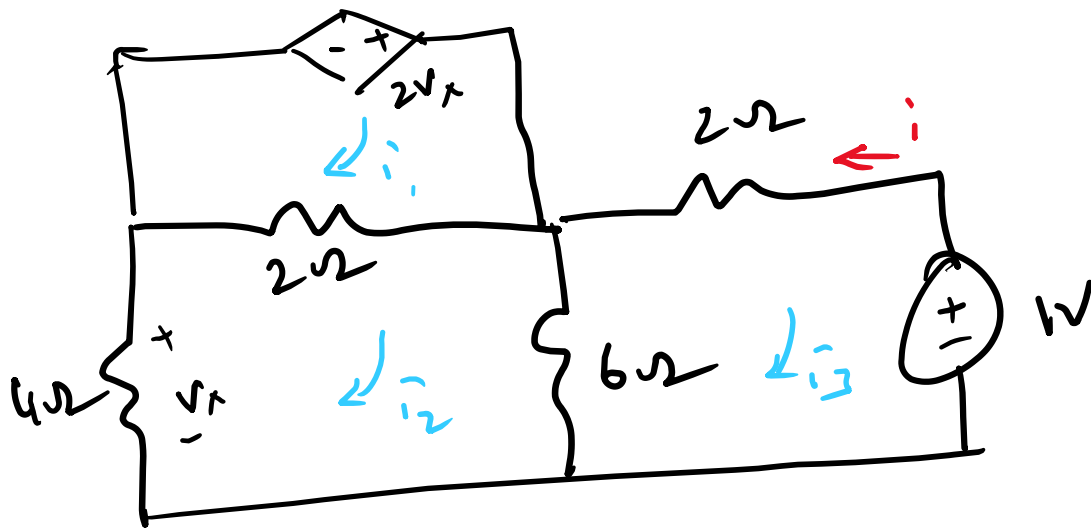
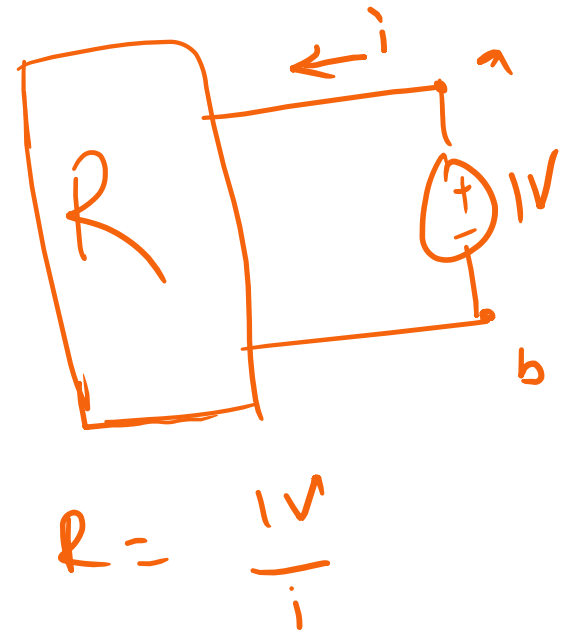
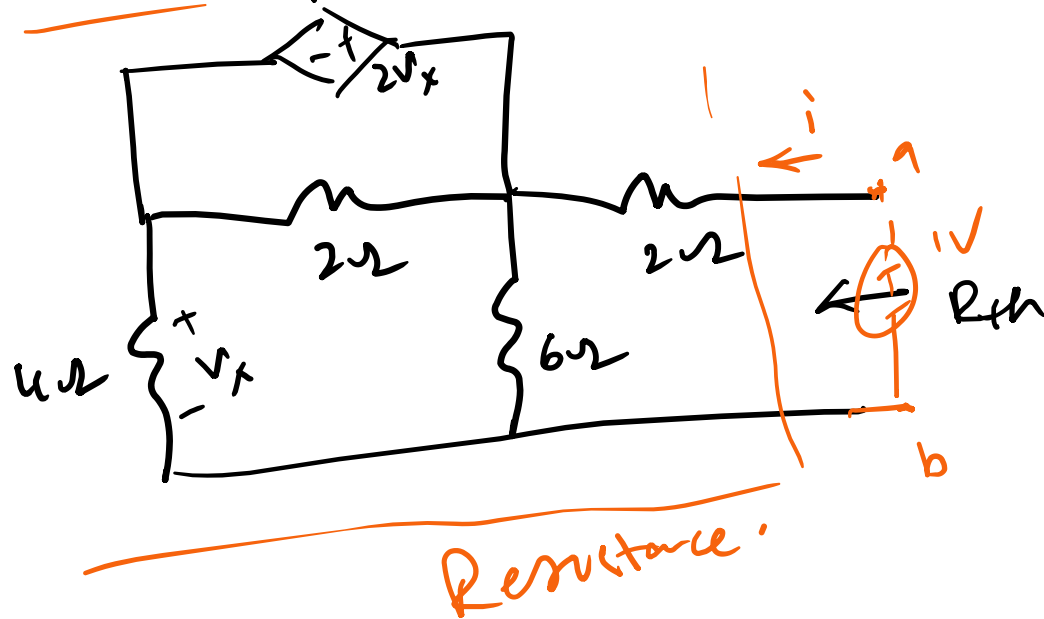
$$V_{oc} \text{ or } v_{th} = \underline{\underline{20V}}$$

$$V_{oc} = \hat{i}_3 \cdot 6$$

$$\hat{i}_1 = 5A$$

$$v_x = 4(\hat{i}_1 - \hat{i}_3)$$

R_{th} : Turn off all independent sources.



$$\hat{i}_1 = -\hat{i}_3$$

$$V_x = 4(-\hat{i}_2)$$

$$\hat{i}_3 = -\frac{1}{6}A$$

$$R_{th} = 6\Omega$$

Thevenin equivalent circuit.

